# A Hundred Years of Business Cycles and the Phillips Curve

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# Motivation

- **Phillips curve** declared **dead**
  - Killed by policy or by luck
  - Non-linear, unstable

- **Aggregate evidence** is **inconclusive**
  - No forecasting performance (Stock and Watson, 2008)
  - Weak identification (Mavroeidis, Plagborg-Møller and Stock, 2014)

- **Cross-sectional evidence**: **flat**
  - Estimated without using aggregate variation (Hazell et al., 2022)

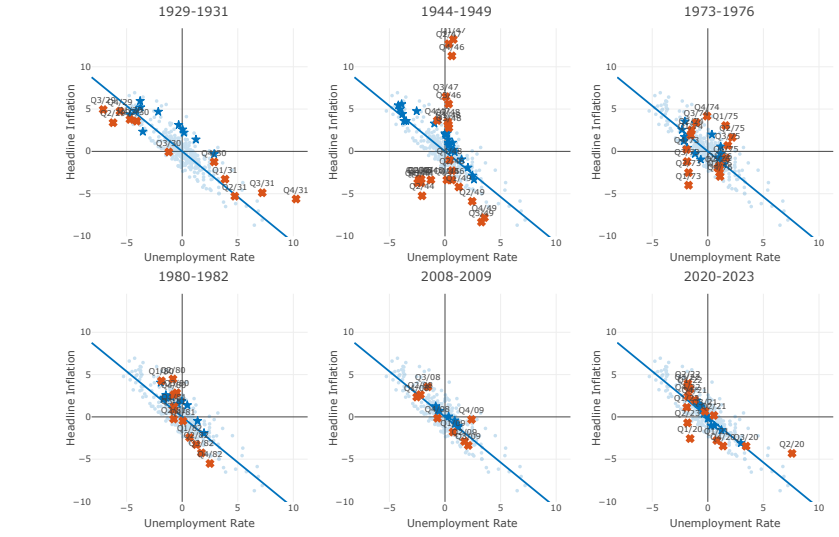
# This Paper

- **Intuition**: raw data are **trend + cycle + noise**

- **Reduced-form TC model** consistent with a **NK economy** with stochastic trends
  - Idiosyncratic trends + one **common trend inflation**
  - One **common business cycle component**
  - Noise absorbed by idiosyncratic cyclical components

- **Long view**: seven quarterly US series from **1900**
  - Output, unemployment, headline and core CPI inflation, SPF and UoM expectations, policy rate

# Trends and Noise Hide the Cycle




**Raw data** vs **cycles**: once trend and noise are removed, **steep Phillips curve** at business cycle frequency

# Results

- A **single common cycle** drives US fluctuations, and it looks like a **demand cycle**
  - Matches NBER dates and the CBO and Greenbook output gaps
- The **policy rate** is mainly driven by the common cycle: **77%** of its cyclical variation
  - Idiosyncratic component small, mainly during the **Volcker disinflation**
- **Reduced-form Phillips curve alive and well**: $-1.08$ over the century
  - $-0.40$ in the raw data
  - $-0.77$ over **1946–84**, $-0.86$ over **1985–2019**
  - Trend inflation is key: holding it constant makes the curve unstable
- **Okun's law** is **at least as stable**: slope $-0.417$
  - $-0.33$ over **1946–84**, $-0.52$ over **1985–2019**
  - Ball, Leigh and Loungani (2017): $-0.421$

# A Stylised NK Model with Trends

# Trend Decomposition

Denote by $\widehat{x}_t$ the **cyclical component** of a variable $x_t$, and by $\tau_t^x$ its **trend**

- Output: $y_t = \tau_t^y + \widehat{y}_t$, **potential output** $\tau_t^y$ is I(2) with a stochastic growth rate
- Unemployment: $u_t = \tau_t^u + \widehat{u}_t$, the **NAIRU** $\tau_t^u$ is a driftless random walk
  - Cycle connected to output via **Okun's law**: $\widehat{u}_t = -\delta_u(L)\,\widehat{y}_t$
- Inflation: $\pi_t = \tau_t^\pi + \widehat{\pi}_t$, trend inflation $\tau_t^\pi$ is a driftless random walk
  - **Inflation expectations**: $E_t\pi_{t+1} = \tau_t^\pi + E_t\widehat{\pi}_{t+1}$, by the martingale property
  - Iterating forward: $\tau_t^\pi = \lim_{h\to\infty} E_t\pi_{t+h}$
- **Policy rate**: $i_t = \tau_t^r + \tau_t^\pi + \widehat{i}_t$, neutral real rate + trend inflation + cycle

# Three-Equation Model and Solution

- **Three-equation model** in the **cyclical components**

$$\widehat{\pi}_t = \beta E_t \widehat{\pi}_{t+1} + \kappa\, \widehat{y}_t + u_t^s$$
$$\widehat{y}_t = \alpha \widehat{y}_{t-1} + (1-\alpha) E_t \widehat{y}_{t+1} - \sigma\left(\widehat{\imath}_t - E_t \widehat{\pi}_{t+1} - u_t^d\right)$$
$$\widehat{\imath}_t = E_t \widehat{\pi}_{t+1} + \theta_\pi \widehat{\pi}_t + \theta_d u_t^d + u_t^{mp}$$

- **Solution**: a **single cyclical state**, $\widehat{y}_t$, determines every cyclical component
  - Unique bounded equilibrium for $\alpha > 0$, $\kappa > 0$ and $\theta_\pi > 0$
  - $\widehat{y}_{t-1}$ is the only endogenous predetermined variable

# Trend-Cycle Representation

- **Observables** admit a **trend-cycle representation**

$$\begin{pmatrix} y_t \\ u_t \\ \pi_t \\ E_t\pi_{t+h} \\ i_t \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \tau_t^y \\ \tau_t^u \\ \tau_t^\pi \\ \tau_t^r \end{pmatrix}}_{\text{trends}} + \underbrace{\begin{pmatrix} 1 \\ -\delta_u(L) \\ \boldsymbol{\Phi} \\ \gamma^h\Phi \\ \Phi(\gamma+\theta_\pi) \end{pmatrix} \widehat{y}_t}_{\text{common cycle}} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ u_t^s \\ 0 \\ \theta_\pi u_t^s + \theta_d u_t^d + u_t^{mp} \end{pmatrix}}_{\text{wedges}}$$

- **Reduced-form Phillips curve** at business cycle frequency: $\widehat{\pi}_t = \boldsymbol{\Phi}\, \widehat{y}_t + u_t^s$
  - $\Phi \equiv \kappa/(1-\beta\gamma)$: reduced-form slope
  - $\gamma \in (0,1)$: persistence of the business cycle, the stable root of the rational expectations system

# An Empirical Model of Trends and Cycles

# Observation Equations

$$\begin{pmatrix} y_t \\ u_t \\ \pi_t \\ \pi_t^c \\ F_t^{spf}\pi_{t+4} \\ F_t^{uom}\pi_{t+4} \\ i_t \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tau_t^y \\ \tau_t^u \\ \tau_t^\pi \\ \mu_t^{spf,\pi} \\ \mu_t^{uom,\pi} \\ \tau_t^r \end{pmatrix}}_{\text{Trends \& biases}} + \underbrace{\begin{pmatrix} 1 \\ \sum_{j=0}^{1}\gamma_{2,j}L^j \\ \sum_{j=0}^{1}\gamma_{3,j}L^j \\ \sum_{j=0}^{1}\gamma_{4,j}L^j \\ \sum_{j=0}^{2}\gamma_{5,j}L^j \\ \sum_{j=0}^{2}\gamma_{6,j}L^j \\ \sum_{j=0}^{2}\gamma_{7,j}L^j \end{pmatrix} \widehat{y}_t}_{\text{Common cycle}} + \underbrace{\begin{pmatrix} \psi_{1,t} \\ \psi_{2,t} \\ \psi_{3,t} \\ \psi_{4,t} \\ \psi_{5,t} \\ \psi_{6,t} \\ \psi_{7,t} \end{pmatrix}}_{\text{Idiosyncratic cycles}}$$

- **Trends**: zero-one loadings, **common trend inflation**, **survey biases**
- **Common business cycle component**: loading on output set to **one**, others free in sign
  - Lead-lag dynamics through the lag polynomials $\sum_j \gamma_{i,j} L^j$
- Idiosyncratic cycles: data-theory **wedges** (measurement error, misspecification)

# Transition Equations

- **Potential output** is **I(2)**: trend growth is a driftless random walk

$$\tau_t^y = \tau_{t-1}^y + g_t^y, \qquad g_t^y = g_{t-1}^y + u_t^g$$

- **All other trends and biases** are **I(1)** driftless random walks

$$\tau_t^x = \tau_{t-1}^x + u_t^{\tau,x}, \qquad x \in \{u, \pi, r\}, \qquad \mu_t^{j,\pi} = \mu_{t-1}^{j,\pi} + u_t^{\mu,j}, \qquad j \in \{spf, uom\}$$

- **Cyclical components**: **ARMA(2,1)** à la Harvey (1985)
  - Simplest process with pseudo-cyclical behaviour: frequency $\lambda$ and damping $\rho$ estimated

$$\begin{pmatrix} \widehat{y}_t \\ \widehat{y}_t^* \end{pmatrix} = \rho \begin{pmatrix} \cos\lambda & \sin\lambda \\ -\sin\lambda & \cos\lambda \end{pmatrix} \begin{pmatrix} \widehat{y}_{t-1} \\ \widehat{y}_{t-1}^* \end{pmatrix} + \begin{pmatrix} v_t \\ v_t^* \end{pmatrix}$$

- Each stationary and non-stationary component is **orthogonal**

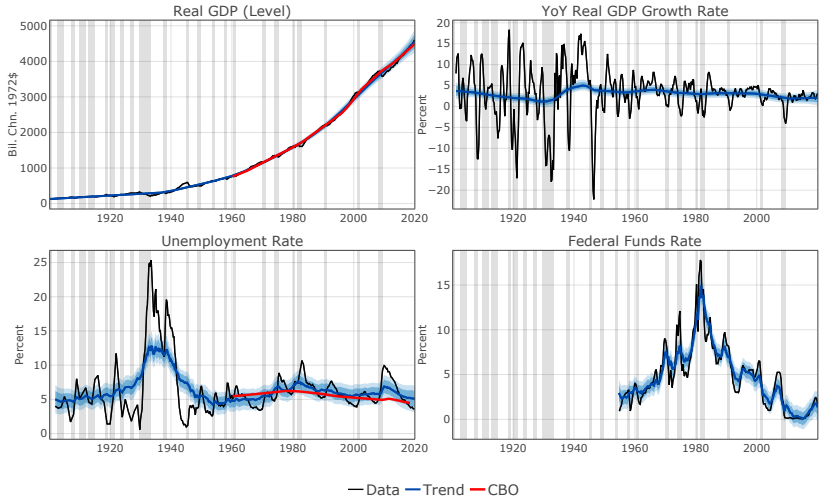
# Data and Estimation

- **Bayesian estimation**: adaptive Metropolis-within-Gibbs and simulation smoother
- **Loose priors**: looser than Bianchi, Nicolò and Song (2026) and than the HP filter ▸ Priors

| Variable | Transf. | Frequency | Period | Source |
|---|---|---|---|---|
| Real GDP | Log-levels | Q | 1900–2019 | Gordon (2007) NBER tables; FRED |
| Unemployment rate | Levels | A1901, Q1929 | 1901–2019 | Haver Analytics |
| Inflation | YoY | A1914, Q1921 | 1914–2019 | Haver Analytics |
| Core inflation | YoY | Q1957 | 1957–2019 | FRED |
| Consumers' exp. inflation | Levels | Q1978 | 1978–2019 | University of Michigan |
| SPF exp. inflation | Levels | S1946, Q1984 | 1946–2019 | Livingston Survey; SPF |
| Short-term interest rate | Levels | Q1954 | 1954–2019 | FRED |

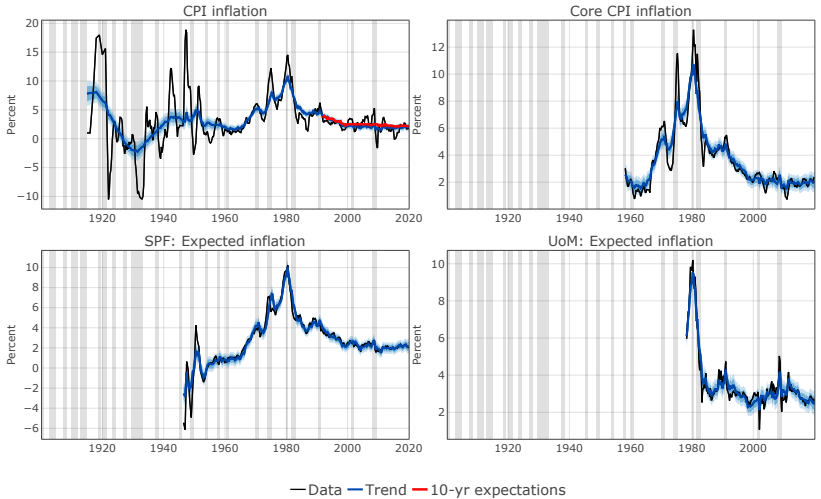
# Trends and Cycles in the US Economy

# Long Run Components (I/II)



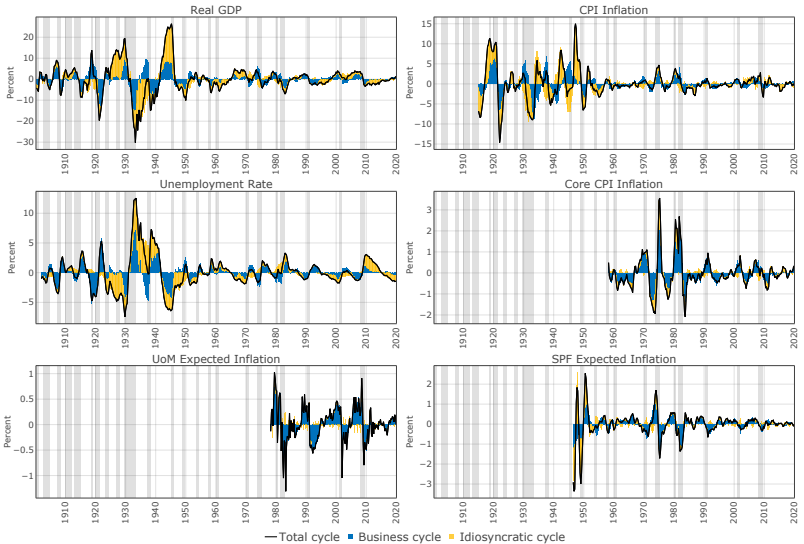

- **Potential output tracks the CBO**; trend growth slows from the 1970s

# Long Run Components (II/II)

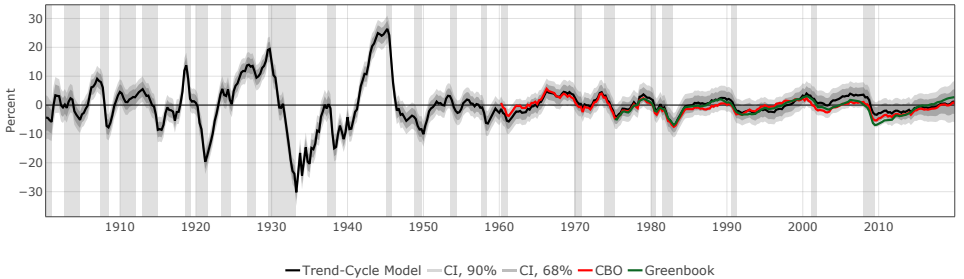



- **Trend inflation tracks the 10-year expectation**, which the model never sees

# Common and Idiosyncratic Cycles

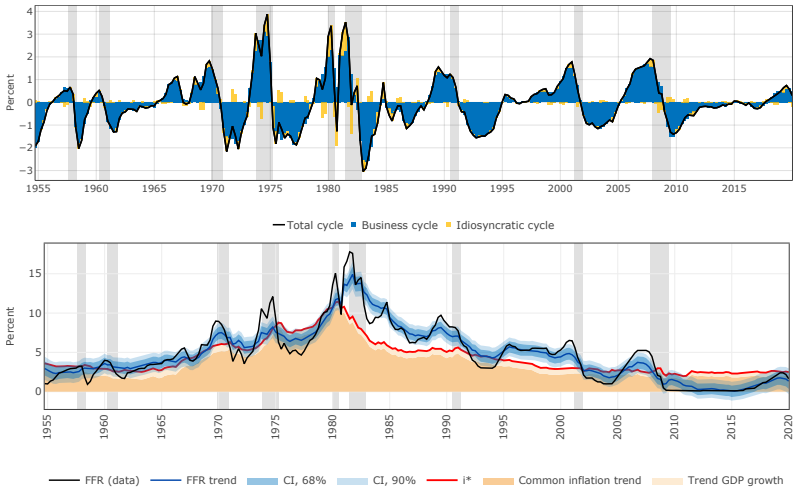


# The Output Gap




- **Model-based output gap**: $y_t - \tau_t^y = \widehat{y}_t + \psi_{1,t}$
  - In percentage deviation from potential output
  - Sum of the common business cycle component and the idiosyncratic cycle of output
- Lines up with **NBER dates** and with the **CBO and Greenbook** gaps until the early 2000s

# The Policy Rate: Cycle and Trend




- Policy-rate trend vs **natural nominal rate** $i_t^* \approx \tau_t^\pi + g_t$: the gap signals the stance

# A Common Source of Business Cycle Fluctuations

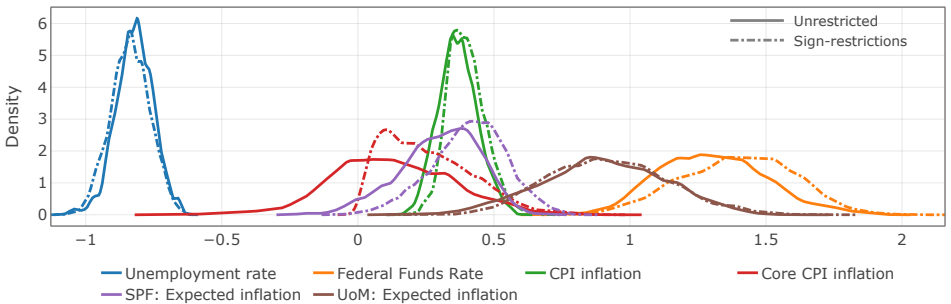
- The **common cycle** explains a **large share** of the cyclical variation of every series
  - Lower share for output, unemployment and CPI inflation: idiosyncratic movements before WWII
  - **77%** of the cyclical variation of the interest rate ▸ Policy rule
  - Robust when the model is fully re-estimated on subsamples ▸ Charts
  - Robust to correlated shocks with trend inflation, NAIRU and trend growth ▸ Charts
- **Not mechanical**: identified by comovement, not by maximising one variance (max share)

COMMON BUSINESS CYCLE SHARE OF THE CYCLICAL DECOMPOSITION (%)

| $y_t$ | $u_t$ | $\pi_t$ | $\pi_t^c$ | $F_t^{spf}\pi_{t+4}$ | $F_t^{uom}\pi_{t+4}$ | $i_t$ |
|---|---|---|---|---|---|---|
| 41.8 | 46.1 | 45.2 | 61.1 | 50.5 | 65.3 | **77.0** |
| [36.4, 47.7] | [40.3, 52.9] | [39.2, 50.0] | [43.8, 73.2] | [38.0, 62.0] | [50.0, 74.8] | [71.8, 81.0] |

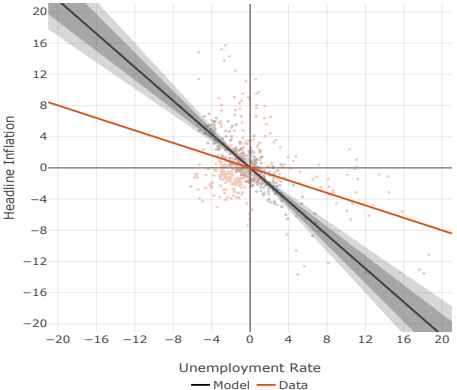
# A Demand Cycle

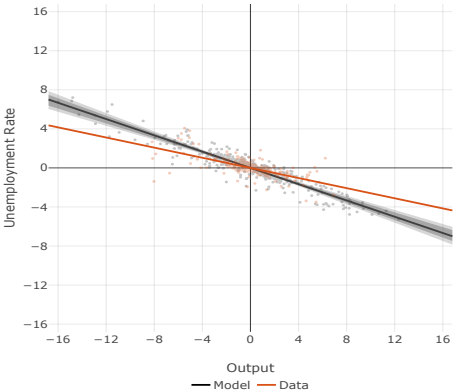
- The common cycle is a **demand cycle**: unemployment falls, prices and the policy rate rise
- Loadings are **not restricted** in sign: imposing sign restrictions only moves core inflation



# Phillips Curve and Okun's Law

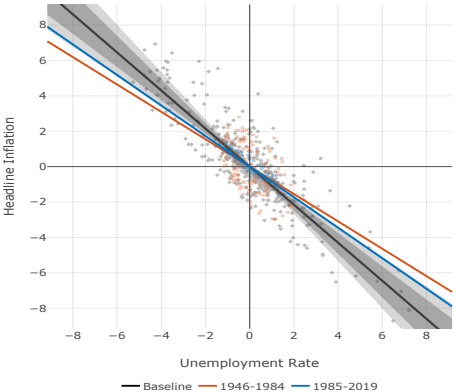
# Phillips Curve and Okun's Law over 120 Years of Data

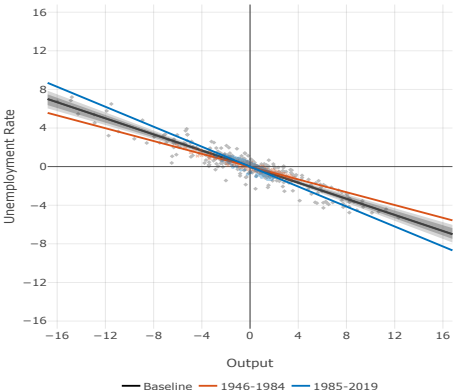



- **Regression coefficient** of the business cycle component of one variable on that of another
- **Phillips curve**: **−1.08** (−0.40 raw data)
- **Okun's law**: **−0.417** (−0.259 raw data)

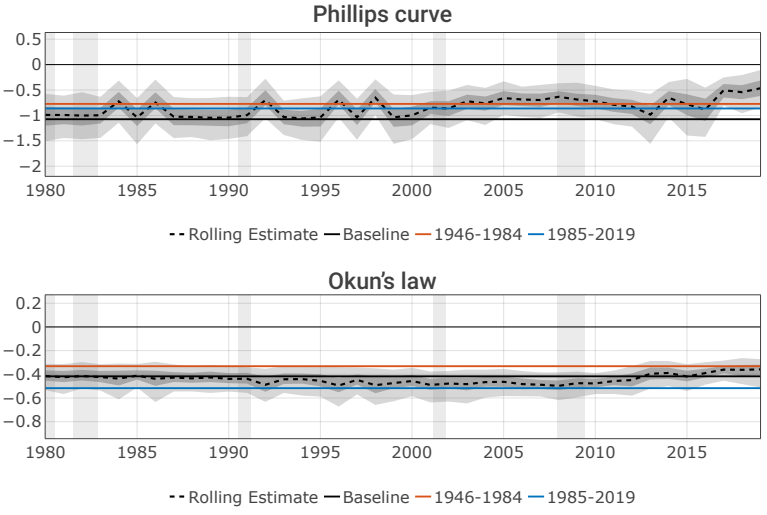
# Stability Across Subsamples




- Model **fully re-estimated** on each subsample, then regressions at business cycle frequency
- **Phillips curve**: **−0.77** over 1946–84, **−0.86** over 1985–2019
- **Okun's law**: **−0.33** over 1946–84, **−0.52** over 1985–2019

# Rolling 80-Year Windows



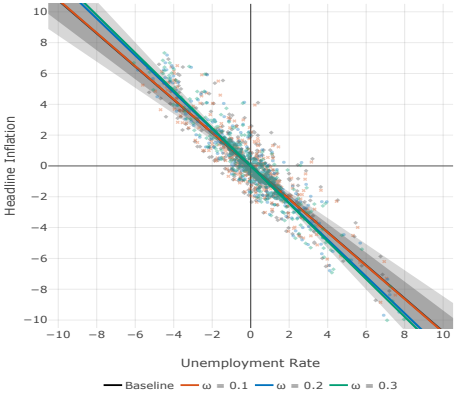

- Model fully re-estimated on **rolling 80-year windows**, ending from 1980 to 2019

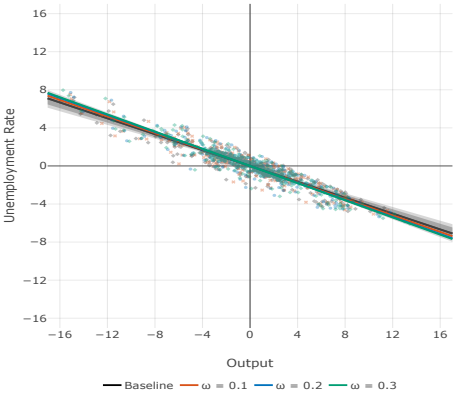
# Trend Inflation Is the Key




- Hold **trend inflation constant**: the curve **flattens**, turns **positive**, and then **almost vertical**

# Robustness: Correlated Trend and Cycle Shocks




- Relax **orthogonality** of cycle and trend shocks (inflation, NAIRU, growth): prior scale $\omega$
  - Prior sd of the cycle–trend correlation: 0.10, 0.18 and 0.25 for $\omega = 0.1, 0.2, 0.3$
- **Both slopes steepen** slightly: the baseline is, if anything, conservative

# Conclusions

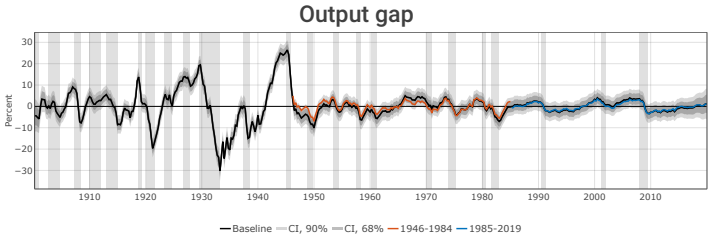
# Conclusions

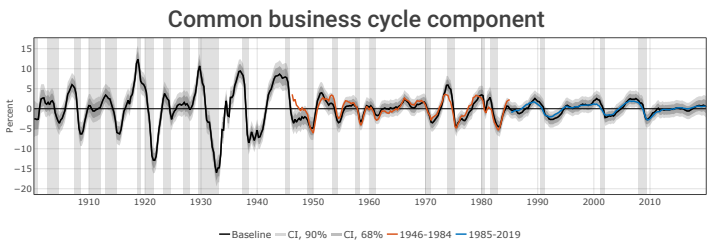
- A **single common cycle**, **demand-like**, drives a century of US fluctuations
  - Policy rate tightly connected to it: idiosyncratic movements mostly during the Volcker disinflation

- At business cycle frequency the **Phillips curve** is **steep and stable**
  - Robust across subsamples and across specifications
  - Controlling for trend inflation is essential: with a constant trend the curve disappears
  - **Okun's law** is at least as stable

# Appendix

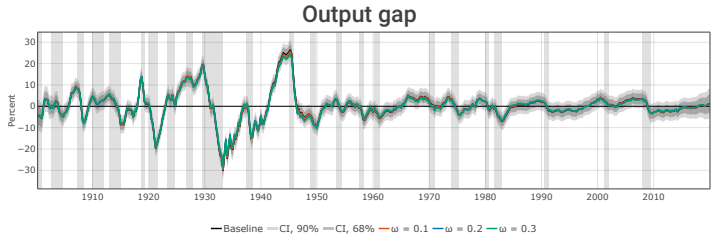
# Subsample Estimates

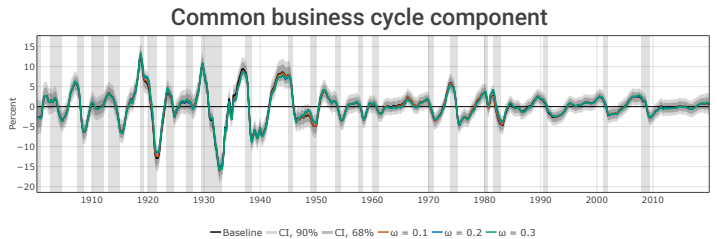



- Full sample against the two postwar subsamples, each fully re-estimated Back

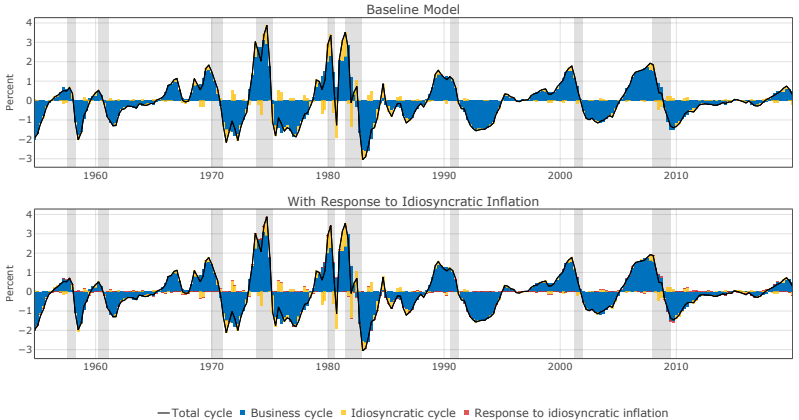
# Correlated Trend–Cycle Shocks




- Baseline (orthogonal) against $\omega = 0.1, 0.2, 0.3$ Back

# Policy Rate and Idiosyncratic Inflation




- Policy rate also responds to the **idiosyncratic inflation cycle**: $\widehat{i}_t = \gamma(L)\widehat{y}_t + \theta_\pi \psi_{3,t} + \psi_{7,t}$
  - Taylor rule: tests whether the Fed responds to inflation off the demand cycle, e.g. supply shocks
- Response **small and short-lived**: the Fed largely looks through it Back

# Prior Distributions

| Name | Support | Density | Parameter 1 | Parameter 2 |
|---|---|---|---|---|
| $\gamma$ (loadings) | $\mathbb{R}$ | Normal | 0 | 1000 |
| $\tau_0$ (initial trend levels) | $\mathbb{R}$ | Normal | $y_0$ | 1000 |
| $\sigma^2$ (shock variances) | $(0, \infty)$ | Inverse-Gamma | 3 | 1 |
| $\sigma_g^2$ (trend-growth shock) | $(0, \infty)$ | Inverse-Gamma | 3 | 0.005 |
| $\rho$ (damping) | $[0.001,\ 0.97]$ | Uniform | 0.001 | 0.97 |
| $\lambda$ (frequency) | $[0.001,\ \pi]$ | Uniform | 0.001 | $\pi$ |

- **Loose trend priors**: implied sd of the expected change in each trend Back
  - About 5.3% over 40 years for trend GDP growth, 6.6% over 20 years for the NAIRU
  - 6–8% over 20 years for trend real rate, trend inflation and survey biases (Bianchi et al.: 1%)
- **1946–84**: **training-sample prior** on the cycle, from a GDP, U, CPI model over 1900–45
  - Surveys sparse before 1984 (Livingston semi-annual, Michigan from 1978): excluded
  - Until core starts in 1957, trend inflation rests on headline CPI alone
  - CPI cycle distorted by price controls after WWII and during the Korean War